

Objective

To analyze cancer incidence and mortality in AYAs diagnosed with cancer between 15 to 39 years old through

- patient demographic characteristics,
- frequencies of major cancer types by sex, and
- trends over time.

Methods

Design

- Retrospective Longitudinal population -based cohort study

Data

- BC Cancer Registry

Outcome

- Counts (Incidence, mortality) & Ratio (M:F)
- Age-standardized incidence rate (ASIR)
- Age-standardized mortality rate (ASMR)

Statistical Analysis

- Descriptive Statistics
- Trends Analysis (join-point regression):
- AAPC (Average Annual Percentage change)
- APC (Annual Percentage Change)
- P<0.05 (Statistically significant)

Flowchart

CAYACS II

(Childhood, Adolescent and Young Adult Cancer Survivors Program)

Diagnosis Year: 1971-2020

Age Group: 15-39 Years

Including more than one disease site at incidence and death date

Incidence: 43,719 cases

Mortality: 6,860 cases

Incidence: 43,588 patients

Mortality: 6,825 patients

58.8%

Female

40.1%

35-39
years

23.4%

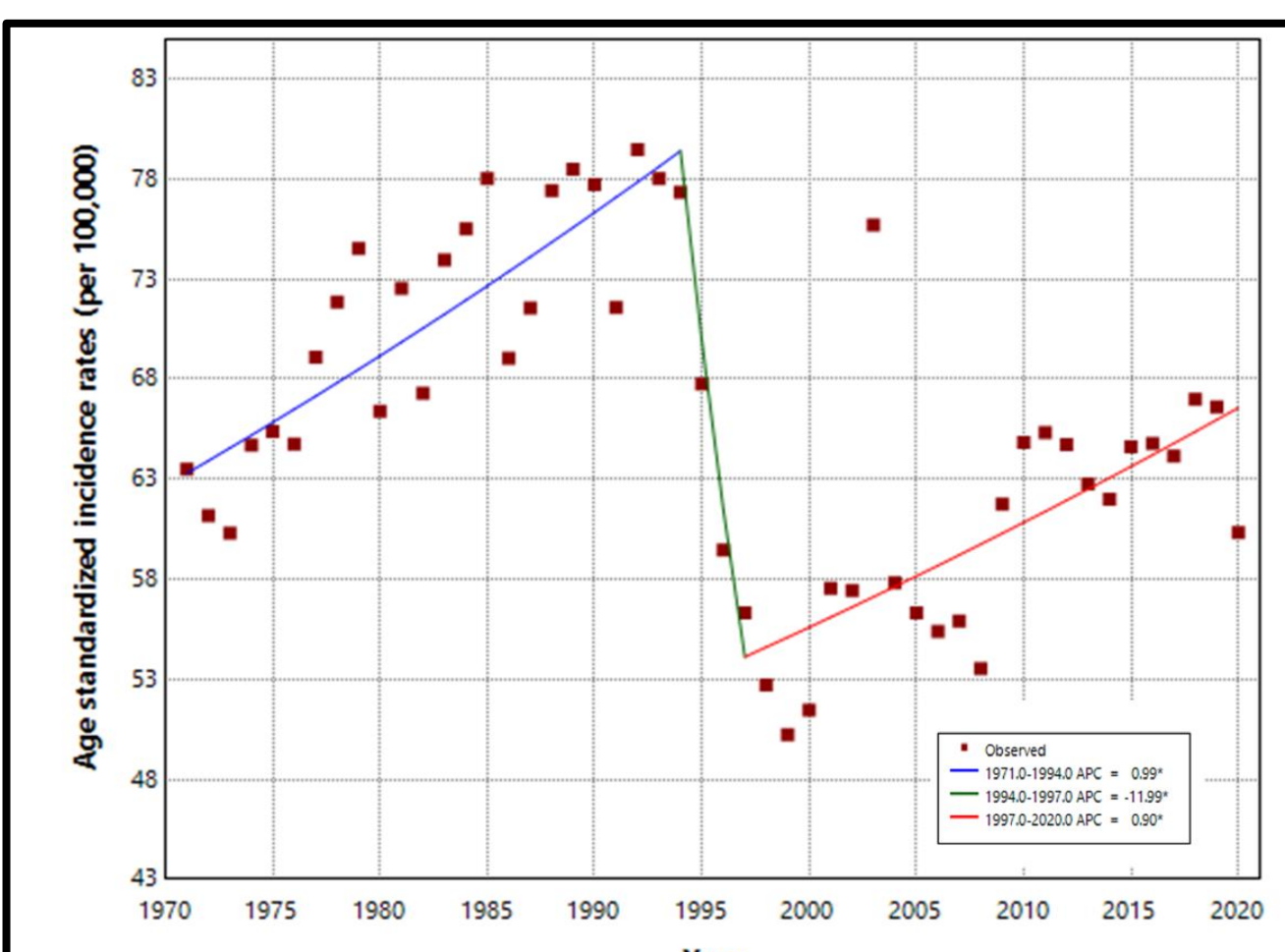
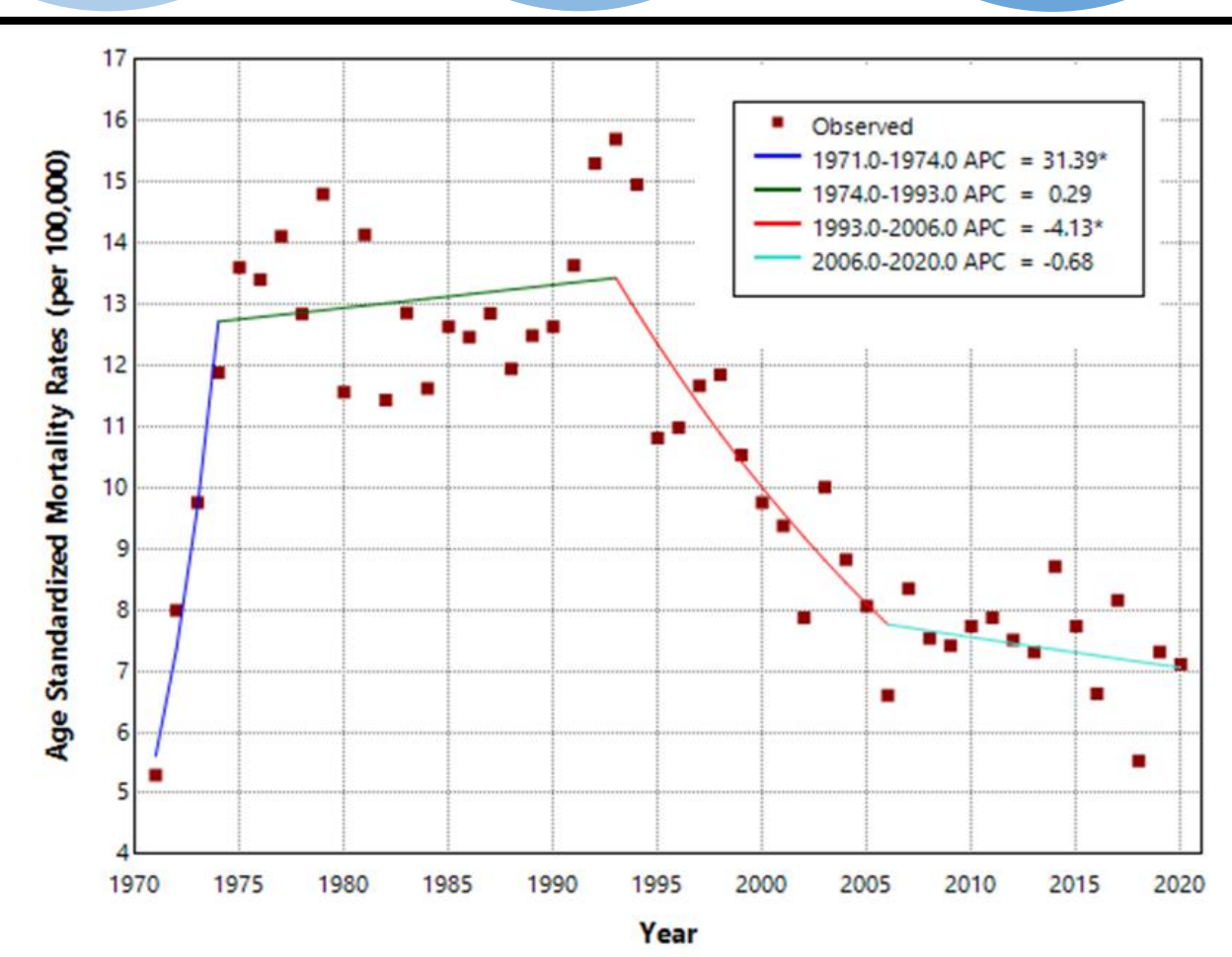
From 2011-
2020

Figure 1. Trends in ASIR and ASMR (all malignancies combined)



* Indicates that the Annual Percent Change (APC) is significantly different from zero at the alpha = 0.05 level. Final Selected Model: 2 Joinpoints.

Incidence

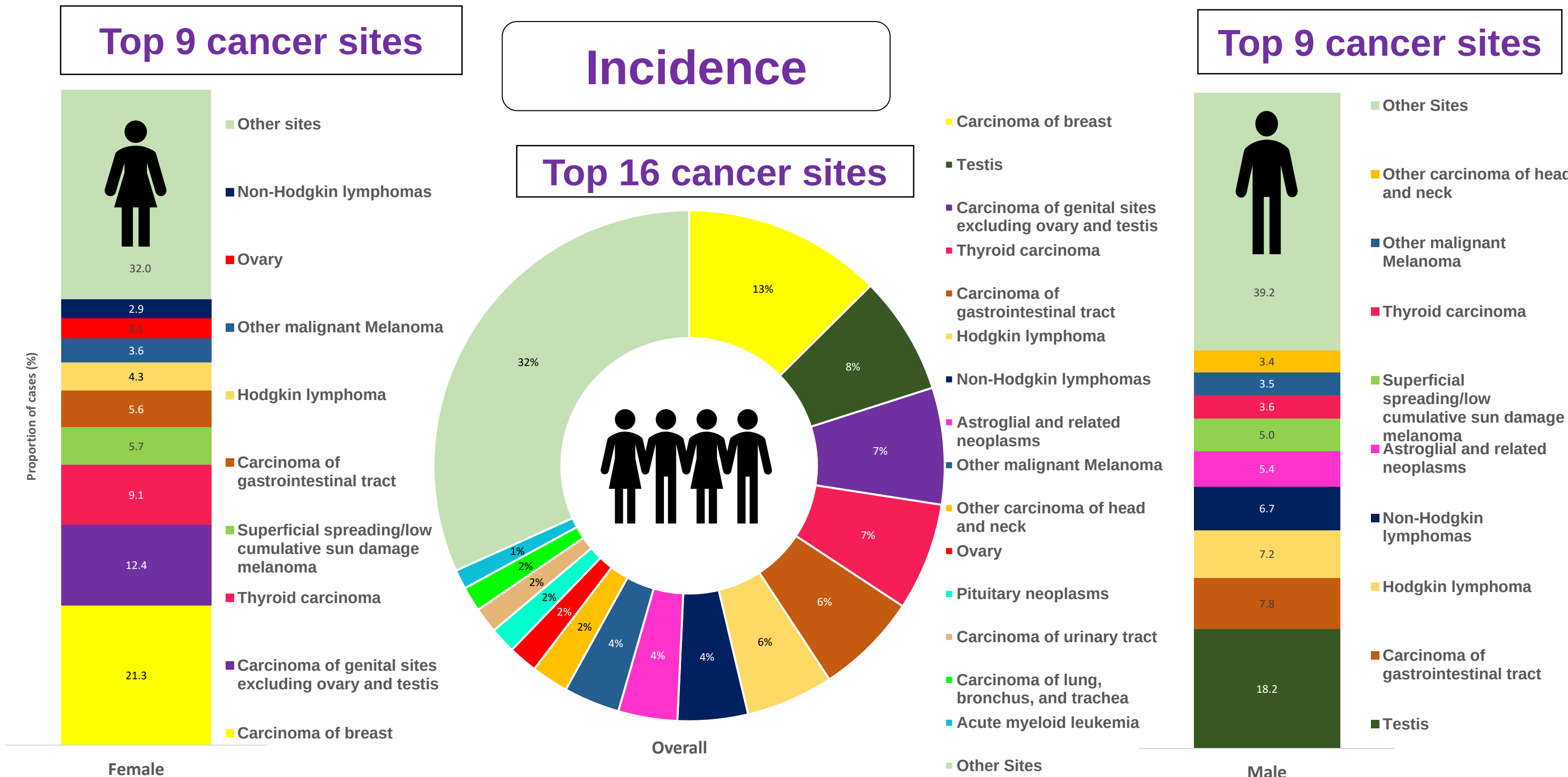


Figure 2. Case distribution of leading cancer type in AYA incidence and mortality.

Trends in incidence

- Overall, **Incidence is rising** for all cancers combined.
- Significant ASIR observed for **thyroid and gastrointestinal cancer**, overall, and for both sexes.
- Among **female**, ASIR for both **Hodgkins and Non-Hodgkins lymphoma** is rising.
- Among **male**, **testicular cancer** has a growing incidence trend.

Join-point analysis (1971-2020)

Cancer Subgroup	AAPC (95% CI)	Trend 1		Trend 2		Trend 3	
		Period	APC (95% CI)	Period	APC (95% CI)	Period	APC (95% CI)
Carcinoma of breast	0.2* (0.0, 0.4)	1971-2020	0.2* (0.0, 0.4)				
Testis	2.4* (0.8, 4.1)	1971-1993	25.0 (-15.5, 84.9)	1973-2020	1.6* (1.3, 1.9)		
Carcinoma of genital sites excluding ovary and testis	0.0 (-1.0, 1.0)	1971-1977	12.1* (6.2, 18.4)	1977-1984	-11.1* (-15.9, -6.2)	1984-2020	0.4* (0, 0.7)
Thyroid carcinoma	2.1* (1.2, 3.0)	1971-1983	-0.6 (-3.6, 2.6)	1983-2020	3.0* (2.4, 3.6)		
Carcinoma of gastrointestinal tract	0.6 (-0.1, 1.2)	1971-1988	-2.0* (-3.5, -0.5)	1988-2020	1.9* (1.3, 2.6)		
Hodgkin lymphoma	0.3* (0.0, 0.6)	1971-2020	0.3* (0.0, 0.6)				
Superficial spreading/low cumulative sun damage melanoma	5.7* (4.3, 7.2)	1971-1981	45.1* (36.2, 54.4)	1981-2020	-2.5* (-3.3, -1.7)		
Non-Hodgkin lymphomas	1.3* (0.5, 2.0)	1971-1994	2.5* (1.4, 3.7)	1994-2020	0.1 (-0.8, 1.1)		
Astroglial and related neoplasms	0.3 (-0.1, 0.8)	1971-2020	0.3 (-0.1, 0.8)				
Other malignant Melanoma	-1.4* (-1.9, -0.8)	1971-2020	-1.4* (-1.9, -0.8)				
Other carcinoma of head and neck	-0.4 (-0.9, 0)	1971-2020	-0.4 (-0.9, 0)				
Ovary	-1.1* (-1.6, -0.6)	1971-2020	-1.1* (-1.6, -0.6)				
Pituitary neoplasms	4.3 (-5.7, 15.4)	1971-1999	-2.9* (-5.5, -0.3)	1999-2002	88.5 (-63.7, 878.7)	2002-2020	5.8* (0.4, 11.6)
Carcinoma of urinary tract	-0.8 (-3.0, 1.4)	1971-1976	-16.3 (-32.1, 3.3)	1976-2020	1.1* (0.3, 1.9)		
Carcinoma of lung, bronchus, and trachea	-0.8 (-2.8, 1.3)	1971-1996	0.2 (-1.3, 1.7)	1996-2004	-11.0* (-19.8, -1.2)	2004-2020	3.2* (0.2, 6.4)
Acute myeloid leukemia	1.2* (0.4, 1.9)	1971-2020	1.2* (0.4, 1.9)				

Table 1. Join point analysis of ASIR per 100 000, AYAs aged 15–39 years in BC

Trends in Mortality

- All-cause mortality from major cancer subgroups is significantly decreasing.**
- An overall **increasing trend** of ASMR was noted for **gastrointestinal cancers in males**.

Join-point analysis (1971-2020)

Cancer Subgroup	AAPC (95% CI)	Trend 1		Trend 2		Trend 3	
		Period	APC (95% CI)	Period	APC (95% CI)	Period	APC (95% CI)
Carcinoma of gastrointestinal tract	4.2* (1.7, 6.8)	1971-1973	230.2* (86.0, 486.3)	1973-1989	-3.7* (-6.0, -1.3)	1989-2020	0.7 (-0.1, 1.5)
Carcinoma of breast	3.6* (1.4, 5.8)	1971-1975	98.9* (53.3, 158.1)	1975-2020	-2.3* (-2.9, -1.6)		
Astroglial and related neoplasms	1.3 (-0.7, 3.3)	1971-1976	23.4* (2.3, 48.7)	1976-2020	-1.0* (-1.7, -0.3)		
Non-Hodgkin lymphomas^	-2.2* (-3.9, -0.4)	1972-1993	1.9 (-1.2, 5.1)	1993-2020	-5.3* (-7.3, -3.2)		
Carcinoma of lung, bronchus, and trachea^^	-3.0* (-5.1, -0.8)	1972-1992	1.3 (-2.2, 4.8)	1992-2014	-6.8* (-9.8, -3.8)		
Carcinoma of genital sites excluding ovary and testis	-0.4 (-1.4, 0.6)	1971-2020	-0.4 (-1.4, 0.6)				
Acute myeloid leukemia	-1.2* (-2.3, -0.2)	1971-2020	-1.2* (-2.3, -0.2)				
Other invasive carcinomas	-0.9 (-2.4, 0.5)	1971-2020	-0.9 (-2.4, 0.5)				

Table 2. Join point analysis of ASMR per 100 000, AYAs aged 15–39 years in BC.

Conclusion

Limitations: All-cause mortality is limited to only 15-39 years old;

Trend Analysis did not include zero rates or missing values.

- The persistent sex disparities in cancer incidence highlight the importance of sex-specific research and targeted prevention strategies.
- Continuous surveillance and tailored interventions are crucial for addressing the unique challenges faced by AYAs with cancer.

References:

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McBride ML, Rogers PC, Sheps SB, et al. Childhood, adolescent, and young adult cancer survivors research program of British Columbia: objectives, study design, and cohort characteristics. Pediatric blood & cancer. 2010

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